

Validity of Three-Fluid Plasma Modeling for Alternating-Current Dielectric-Barrier-Discharge Plasma Actuator

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The validity of numerical modeling of discharge plasma in dielectric barrier discharge plasma actuator is evaluated by comparing the simulation and experimental results under the same conditions. Discharge plasma behavior is numerically simulated based on a three-fluid plasma model. Regarding the induced jet structure, although the simulation tends to underestimate the jet thickness and overestimate the maximum velocity, the discrepancies are small considering the quite simple modeling. In addition, the values of thrust force in the simulation are consistent with those in the experiment only within 10–20%. On the other hand, the simulation underestimates the power consumption by 40–50% mainly due to the assumption of two-dimensionality. Furthermore, the thrust force and power consumption are investigated from the viewpoint of characteristics for voltage amplitude, voltage waveform, and dielectric material. As a result, the discharge mode in the simulation, which strongly affects the electrohydrodynamic force generation characteristics, needs to be consistent with that in the experiment to reproduce the voltage waveform characteristics of thrust force. In addition, to reproduce the dielectric material characteristics of thrust force, not only permittivity but also secondary electron emission coefficient needs to be set to appropriate values for each material.

Nomenclature

D_e, D_p, D_n	=	diffusion coefficient for electron, positive ion, and negative ion
E	=	electric field vector
e	=	elementary charge
F	=	body force vector
j_e, j_p, j_n	=	electron, positive ion, and negative ion current density
n_e, n_p, n_n	=	electron, positive ion, and negative ion number density
p	=	ambient gas pressure
r_{ep}	=	recombination coefficient between electron and positive ion
r_{pn}	=	recombination coefficient between positive ion and negative ion
T	=	one period of the alternating current voltage
t	=	time
V	=	electric potential
v_e	=	mean velocity of electron
α	=	ionization coefficient
γ	=	secondary electron emission coefficient
δ_s	=	Dirac delta function
ϵ_0, ϵ_r	=	electric permittivity in the vacuum and relative electric permittivity of the dielectric

η	=	attachment coefficient
μ_e, μ_p, μ_n	=	electron, positive ion, and negative ion mobility
σ	=	surface charge density

Subscripts

e	=	electron
n	=	negative ion
p	=	positive ion

I. Introduction

ACTIVE flow control methods have been intensively studied in order to innovate the performance of fluidic machinery. Among various active flow control methods, plasma-assisted flow control concept attracts a lot of attention in recent years because of advantages it provides, such as simple structure that has no mechanical moving parts and quick response to control input compared with existing flow control devices. Surface dielectric barrier discharge (DBD) plasma actuator is a type of such flow control device, and its feasibility has been intensively studied in the last two decades [1–3]. The typical configuration of DBD plasma actuator is shown in Fig. 1. It consists of a pair of electrodes separated by a dielectric. One electrode is exposed to the flow and the other is covered. Depending on a voltage waveform supplied to the electrodes, the DBD plasma actuators can be mainly divided into an alternating current dielectric barrier discharge (AC-DBD) plasma actuator and nanosecond pulsed dielectric barrier discharge (NS-DBD) plasma actuator.

The dominant mechanism for flow control of AC-DBD plasma actuator is the electrohydrodynamic (EHD) force acting on the ambient gas [1–3]. When an AC high voltage (typically several kilohertz and several kilovolts) is applied, weakly ionized plasma is generated by the discharge around the exposed electrode. The charged particles are accelerated by the electric field and collide with the neutral molecules, resulting in the EHD force. The EHD force induces a surface-jet, which modifies the flowfield around the fluidic machinery by directly injecting the momentum to boundary layer. The AC-DBD plasma actuator has been used in a large variety of applications. The pioneering work by Roth et al. [4] is the suppression

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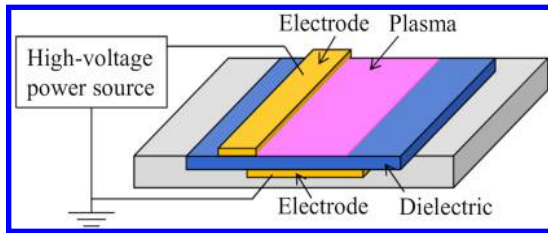


Fig. 1 Schematic of a DBD plasma actuator.

of flow separation over a NACA 0015 airfoil at $Re = 6.4 \times 10^4$, and a number of experimental and computational studies have clearly demonstrated the capability of AC-DBD plasma actuators to manipulate the dynamics of different flows, such as flow separation control [5–7], laminar-to-turbulent transition delay [8,9], vortex shedding control [10], and noise reduction [11]. Some studies have also demonstrated the ability of AC-DBD plasma actuators for film cooling efficiency enhancement [12,13]. However, a survey of available studies indicates that the AC-DBD plasma actuator is applicable mainly to low subsonic flow regime with $U_\infty \leq 30$ m/s and $Re \sim 10^5$ because the induced flow velocity hardly exceeds 10 m/s [3].

On the other hand, the NS-DBD plasma actuator has demonstrated a capability of flow separation control over a wide velocity range [14]. The NS-DBD plasma actuator is driven by repetitive high-voltage pulses with typical duration of a few tens of nanosecond. The fast discharge induces a rapid energy deposition in the discharge layer, resulting in a rapid (typically less than $1 \mu\text{s}$) temperature rise via a Joule heating, which induces a spherical shock wave and density gradient. Thus, in contrast to the AC-DBD, the flow control mechanism of NS-DBD plasma actuator is a Joule heating effect [15]. The NS-DBD plasma actuator has proved to be effective in flow separation control with a flow Reynolds number up to 2.3×10^6 [16–18].

Although the NS-DBD plasma actuator is considered to be superior in terms of the applicability to high main flow velocity, the AC-DBD plasma actuator is also believed to be the promising candidate for flow control device owing to its high applicability to various flow patterns. This study focuses on the AC-DBD plasma actuator (hereinafter, this is called “DBD plasma actuator”).

Many research efforts have been devoted to understanding the physical mechanism in the DBD plasma actuator. In experimental studies, the effects of geometrical and driving conditions, such as applied voltage amplitude, voltage waveform, dielectric material, and dielectric thickness, have been intensively investigated. The results are summarized in the review papers [1,2,19]. However, the detailed plasma dynamics cannot be easily analyzed due to the limitation of experimental methods. Therefore, in addition to the experimental investigations, numerical investigations have provided valuable insights into the physical mechanism. Among numerical models of discharge plasma in the DBD plasma actuator proposed by several research groups [20–26], a numerical modeling based on fluid approximation has been often adopted for numerically simulating the DBD plasma actuator owing to its simple physical model and low computational costs [20,21]. Boeuf et al. firstly conducted numerical simulations using the fluid model with an assumption that the plasma consists of only the electron and single-positive ion in pure nitrogen; however, the discrepancies in the characteristics of EHD force generation were observed [20]. In their later works, using a three-fluid model considering the negative ion species in addition to the simplest model, they have comprehensively analyzed the characteristics of discharge plasma and EHD force generation [21]. They have clarified that the discharge characteristics are different between the positive-going and negative-going voltage phase, and the negative ions generated in the negative-going phase play an important role in the EHD force generation [21]. Similar three-fluid plasma model was adopted by many research groups [27–31], and these previous studies have provided important knowledge about the physics. However, the validity of three-fluid model remains controversial.

In the typical fluid model adopted in many studies, several strong assumptions are used. The local field approximation (LFA) is generally adopted; that is, the swarm parameters, such as the ionization and attachment coefficients, become functions of the local reduced

electric field [20,21,23,25,27–33]. The LFA is generally accepted to be valid at atmospheric pressure because the energy relaxation time of electrons is sufficiently short compared with the discharge plasma except within the regions of high electric field. The simplified Einstein’s relation with a constant electron temperature for the diffusion coefficient and the electrostatic field are also assumed [20,21,25,27–33]. In addition, only several kinds of plasma reaction (typically less than six reactions) are taken into account [20,21,27–33]. Although some research groups investigate the effects of detailed plasma chemistry processes, such as excitation [22,23], photoionization [24], and electromagnetic field [25], the simple model is mainly adopted for simplicity and stable computation. On the other hand, there are severe difficulties also in computation from the viewpoint of numerical cost. The time step on the order of 10^{-13} – 10^{-14} s is required for the stable computation with the explicit method [27,28], and fine grid spacing (less than $2 \mu\text{m}$) is required to resolve the plasma sheath. However, use of such short time stepping and fine grid spacing is impractical because the computation in several AC voltage cycles (hundreds of microseconds) and large spatial scale (several millimeters) is required for the simulation of DBD plasma actuator. Therefore, the two-dimensionality of discharge is assumed, and although large numerical error arises, the implicit method and coarse grid spacing (more than $7 \mu\text{m}$) is frequently used [20,21,29–33]. It has been reported that the simple three-fluid simulation with the implicit method and coarse grid can qualitatively reproduce the behavior of discharge plasma evolution and tendencies of the EHD force generation in spite of the strong assumption and large numerical error [32,33]; however, the simulation results were compared with experimental results using different conditions: for example, the discharge characteristics were compared between simulation results with a voltage amplitude of 20 kV and a dielectric thickness of 0.5 mm and experimental results with a voltage amplitude of 5 kV and a dielectric thickness of 1.8 mm [32]. To the author’s knowledge, simulation results using the fluid model have not been compared quantitatively to experimental results with the same conditions. It is significant for further research and development of plasma actuator to discuss the validity of the simple three-fluid model with practical computational settings, that is, to clarify what characteristics the three-fluid model can and cannot reproduce and what conditions are required to ensure the reliability of the simulation results.

Based on this background, the objective of this study is to evaluate the validity of the simple three-fluid model with practical computational settings for simulating the DBD plasma actuator. We conduct numerical simulation and experiment under the same conditions. The discharge plasma behavior is numerically simulated based on the three-fluid model using the assumptions generally adopted for simulating the DBD plasma actuator, and the simulation is conducted using practical computational settings generally used for analyzing the DBD plasma actuator. First, the jet structure induced by a DBD plasma actuator is compared. Next, the voltage amplitude characteristics of thrust force and power consumption, which are typically used as basic characteristics to indicate the performance of a DBD plasma actuator, are compared. In addition, the thrust force and power consumption are investigated from the viewpoint of characteristics for applied voltage waveform and dielectric material. Finally, we discuss the effects of discharge mode on the reproducibility using the three-fluid model.

II. Simulation Model

The ambient gas is considered to be the atmospheric air in this study. The numerical model is the same as the plasma fluid model used in the Refs. [20,21]. We do not consider the detail plasma chemistry in the air; the electron, one type of positive ion, and one type of negative ion are considered as plasma particles, and electron impact ionizations ($\text{N}_2 + e \rightarrow \text{N}_2^+ + e + e$, $\text{O}_2 + e \rightarrow \text{O}_2^+ + e + e$), electron attachments (three body and dissociative attachment of O_2), and recombination are taken into account as plasma chemistry. The continuity equations of plasma particles with the drift-diffusion flux are solved coupling with the Poisson’s equation for electric potential. The governing equations are as follows:

$$\frac{\partial n_e}{\partial t} + \nabla \cdot (-n_e \mu_e \mathbf{E} - D_e \nabla n_e) = (\alpha - \eta) n_e |\mathbf{v}_e| - r_{ep} n_e n_p \quad (1)$$

$$\frac{\partial n_p}{\partial t} + \nabla \cdot (n_p \mu_p \mathbf{E} - D_p \nabla n_p) = \alpha n_p |\mathbf{v}_e| - r_{ep} n_e n_p - r_{pn} n_p n_n \quad (2)$$

$$\frac{\partial n_n}{\partial t} + \nabla \cdot (-n_n \mu_n \mathbf{E} - D_n \nabla n_n) = \eta n_e |\mathbf{v}_e| - r_{pn} n_p n_n \quad (3)$$

$$\nabla \cdot (-\epsilon_r \nabla V) = \frac{e}{\epsilon_0} (n_p - n_e - n_n) + \frac{e}{\epsilon_0} \sigma \delta_s \quad (4)$$

The surface charge density σ is expressed by the Dirac delta function δ_s , which becomes unity only on the dielectric surface and zero in other area, and it is calculated self-consistently by integrating the electron and ions fluxes to the surface. Electrons and ions are supposed to recombine instantly on each surface element. The swarm parameters (the electron mobility, ionization, and attachment coefficients) are determined with the local electric field approximation (LFA) and calculated by the BOLSIG+ software [34] assuming that the ambient gas is air ($N_2:O_2 = 0.8:0.2$). The ion mobilities are set as constant values: $\mu_p = \mu_n = 26.7/p \text{ [m}^2 \cdot \text{V}^{-1} \cdot \text{s}^{-1}\text{]}$. The diffusion coefficients are determined from the following Einstein relations; $D_e/\mu_e = 1 \text{ V}$ and $D_{pn}/\mu_{pn} = 0.01 \text{ V}$. Although the recombination coefficients are different depending on the reaction process, the coefficient of electron-ion recombination r_{ep} and ion-ion recombination r_{pn} is assumed to be the typical constant value for simplicity: $2 \times 10^{-13} \text{ m}^3 \cdot \text{s}^{-1}$ as in the previous studies [20,21].

The governing equations are solved by the finite volume method; the drift terms in Eqs. (1–3) are evaluated by the upwind scheme with the MUSCL interpolation; the diffusion terms are evaluated by the second-order-accurate central difference scheme; the time integration is conducted by the LU-ADI implicit scheme; the Poisson's equation is solved by the SOR method with semi-implicit technique for the coupling between charged particle transport and Poisson's equation.

The EHD force vector $\mathbf{F}$ in collisional plasma can be obtained assuming that the force is equal to the rate of momentum transfer per unit volume due to the collision between plasma particles and neutral particles:

$$\mathbf{F} = \frac{j_p}{\mu_p} - \frac{j_e}{\mu_e} - \frac{j_n}{\mu_n} \quad (5)$$

$$j_k = \pm en_k \mu_k \mathbf{E} - e D_k \nabla n_k (k = e, p, n) \quad (6)$$

The computational settings for the plasma simulation are shown in Fig. 2. The downstream edge of the top electrode is set at the x -axis origin. In the initial state, the computational domain is filled with uniform plasma ($n_e = n_p = 10^7 \text{ m}^{-3}$) [20]. The alternating voltage is applied to the top electrode, and the lower electrode is electrically

grounded throughout the simulation. We assume a quasi-steady state after applying 4.25 cycles of the alternating voltage, and we calculate the time average over 2 cycles within the quasi-steady state. In Fig. 2, the boundary conditions of the electric potential and plasma number densities are described. For plasma number density, the symmetric boundary conditions are implemented at the left, right, and top side of the domain and the surfaces of top electrode and dielectric. For the electric potential, the symmetric boundary conditions are implemented at the left, right, and top side of the domain. As the Dirichlet boundary conditions for the electric potential, the AC voltage is applied to the top electrode, and the bottom of the domain, which corresponds to the grounded electrode, is electrically grounded. In addition, the secondary electron emission by positive ion bombardment is taken into account at the dielectric and electrode surface; γ is set to be 0.05 as in previous studies [20,21], whereas γ is an electrical physical property value of the dielectric material. The secondary electron emission is considered when positive ions strike on the dielectric surface as

$$j_{e,\perp} = \gamma j_{p,\perp}$$

where $j_{e,\perp}$ and $j_{p,\perp}$ are the perpendicular components of the electron and positive ion current to the dielectric surface, respectively. These boundary conditions are typically implemented in numerical studies of the DBD plasma actuator [20,21,29–33]. Note that the boundary conditions at the left, right, and top of the domain are expected to have few influences on the results because the height and width of air layer are set to be large enough based on our numerical tests.

The simulation domain is divided into rectangular grid system. The number of cells is 750×275 . The grid spacing is constant at $20 \text{ }\mu\text{m}$ in the horizontal direction, and it is gradually stretched from $20 \text{ }\mu\text{m}$ in the vertical direction. Note that the grid spacing is constant at $20 \text{ }\mu\text{m}$ within 1.0 mm height from the dielectric surface. Although such large grid spacing cannot resolve plasma sheath structure, it is typically used for simulating the DBD plasma actuators due to the computational cost [20,21,29–33], and therefore the model validation in the practical computational settings is very valuable. The time-step interval is determined from the CFL condition; the Courant number is set at 5.0.

To obtain the flowfield in the simulation, the incompressible Navier–Stokes simulation is conducted. The time-averaged EHD force distribution is incorporated with the Navier–Stokes equations as the source term. The thrust force, which is defined as the reaction force against the jet generation, is calculated by integrating the flow momentum fluxes into/out the control volume in the time-averaged flowfield. The control volume is set to the area of $x = -2$ to 23 mm and $y = 0$ to 6 mm . The thrust force corresponds to the EHD force produced by the DBD plasma actuator minus the viscous and pressure drag, and typically used to evaluate the performance of the DBD plasma actuator. Note that, in order to estimate the drag of the fluid on plate, the downstream edge of the control volume is determined based on the experimental settings: the length from the edge of top electrode to the edge of dielectric plate is 23 mm . Because the flowfield is

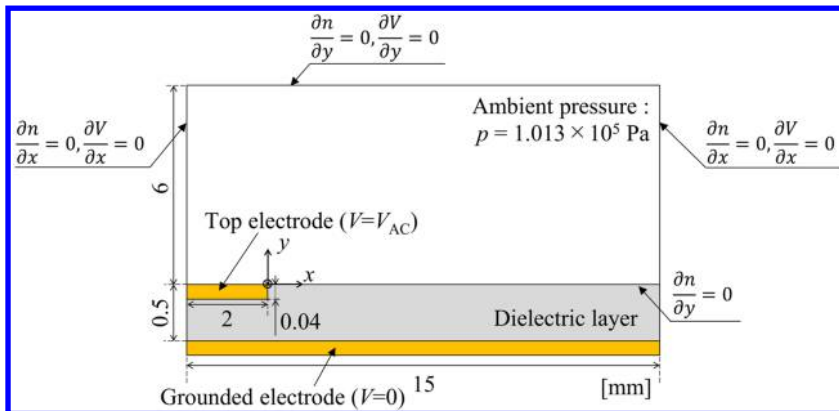


Fig. 2 Simulation domain and settings.

expected to be still laminar in the vicinity of the DBD plasma actuator, the two-dimensional laminar flow is assumed in the fluid simulation. For more information on the numerical model and computational settings for the fluid simulation, refer to the previous study [35].

III. Experimental Settings

In this section, the experimental setup is explained. To quantitatively compare the simulation results with experimental results, we conduct the experiment under the same conditions as in the simulation. The thrust force and power consumption measurements are conducted to evaluate the performance of a DBD plasma actuator, and the PIV measurement is conducted to obtain the flowfield around the actuator.

Actuators are composed of a dielectric plate and two electrodes; the thickness of top and grounded electrode is 0.03- and 0.05-mm-thick copper tapes, respectively. The streamwise width of the top and grounded electrode is 10 and 20 mm, respectively. The spanwise length of the electrodes is 130 mm. The electrodes are arranged with 2 mm overlapping. The dielectric thickness is 0.5 and 1.0 mm. Three types of materials are used for the dielectric: glass epoxy, polyimide, and polytetrafluoroethylene (PTFE). Table 1 shows the physical properties of three materials, especially the electrical characteristics. Note that, although the top electrode is flush-mounted in the dielectric layer in the simulation as shown in Fig. 2, it is not flush-mounted in the experiment. In addition, the length of the top and grounded electrodes in the experiment is different from that in the simulation; the length of the top electrode is 2 and 10 mm, and the length of the grounded electrode is 15 and 20 mm in the simulation and experiment, respectively. However, the difference in the length of the electrodes does not affect the characteristics of discharge plasma and EHD force generation because the discharge plasma propagates downstream from the downstream edge of the top electrode and it stops without reaching the downstream edge of the grounded electrode in both the experiment and simulation.

Figure 3 shows the experimental system for the thrust and the consumption power measurement. The voltage waveform is generated by a function generator (Tektronix AFG3021) and amplified by a high-voltage amplifier (Trek Model 10/40A). The voltage waveforms are sinusoidal, triangular, and square waveforms. The amplitude is in a range from 4 to 10 kV_p, and its frequency is 750 Hz and 5 kHz. In the thrust measurement, the DBD plasma actuator is vertically mounted on a stand put on an electronic balance (SHIMADZU, UW620ZH, resolution: 1 mg, 0.076–0.081 mN/m). The thrust is measured from the weight difference between before and after the actuation. In the power consumption measurement, a monitor capacitor is inserted to the ground line in series (250 nF) [36]. The power consumption is calculated by using V-Q Lissajous figure, which can be obtained by measuring the electrical charge accumulated in the monitor capacitor. Also in the simulation, the power consumption is evaluated by using the Lissajous figure method.

Figure 4 shows the PIV system. The PIV system consists of a double-pulse Nd-YAG laser (Litron Lasers, NanoS30-16PIV) and a complementary metal oxide semiconductor (CMOS) camera with resolution of 2560 × 2160 pixels (Neo5.5). The DBD plasma actuator is placed in a stainless chamber, and tracer particles generated by incense sticks are seeded inside the chamber. A single focus lens (Nikon, Micro-Nikkor 105 mm f/2.8) and a band-pass filter are attached to the CMOS camera. The PIV sampling rate is 9 Hz. The time-averaged flowfield is calculated from 215 instantaneous flowfields. The measurement area is in the vicinity of the electrode edge (37.4 mm × 28.5 mm). The multiple correlation is used as the PIV

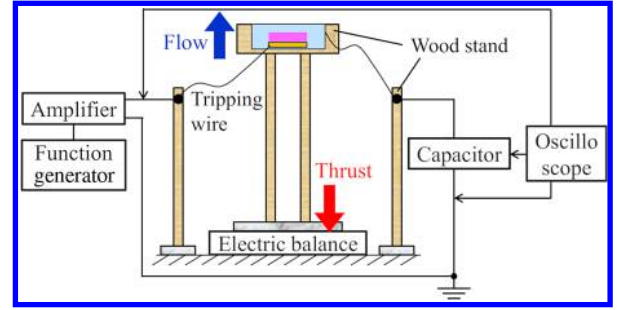


Fig. 3 Experimental setup of thrust force and power consumption measurement.

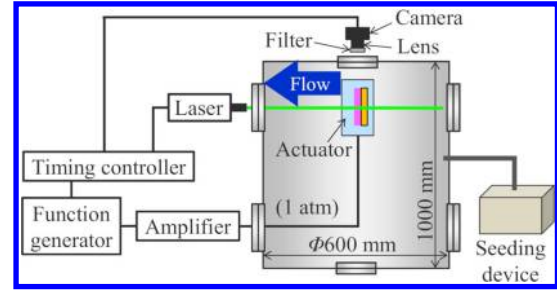


Fig. 4 Experimental setup of PIV measurement system.

algorithm. The interrogation window is 32×32 pixels and its overlap is 50%. The spatial resolution is $230 \mu\text{m}$. The threshold of tracer particle displacement is set to 10 pixels to remove incorrect velocity vectors.

IV. Simulation Results

A. Grid Convergence

In this section, we check the grid convergence of the plasma simulation. The space-integrated value and distribution of the time-averaged EHD force generation are investigated because it is reported that the total induced momentum per unit time and the center of gravity of the EHD force distribution are highly relevant to the flowfield induced by the DBD plasma actuator [37]. In addition, time histories of discharge current and space-integrated EHD force are investigated. The simulation is conducted with different grid spacing: $\sqrt{2}$, 2, and $2\sqrt{2}$ times higher and $\sqrt{2}$ and 2 times lower grid resolution compared with $20 \mu\text{m}$. Note that these grid spacings are relatively large (more than $7 \mu\text{m}$) because the objective is to discuss the validity of the numerical modeling in the practical computational settings. As for finer grid spacing, the grid convergence has been discussed in Ref. [38] for the NS-DBD plasma actuator. The time interval is also changed in response to the grid spacing: $\sqrt{2}$, 2, and $2\sqrt{2}$ times smaller and $\sqrt{2}$ and 2 times larger for the finer and coarser grid, respectively. The applied voltage waveform is a sinusoidal waveform with the amplitude of 15 kV_p and frequency of 20 kHz, and the relative permittivity of dielectric is set to 5.

Figures 5 and 6 show the time histories of discharge current and space-integrated EHD force with different grid spacings, respectively. The horizontal axis shows the dimensionless time t normalized by one period of AC voltage cycle T . Figure 7 plots the time-averaged and space-integrated EHD force as a function of the grid spacing. Figure 8 shows the contours of time-averaged EHD force distributions with different grid spacings. In Fig. 8a, the overlapped contours of the results with grid spacing of 10 and $40 \mu\text{m}$ are shown, and the black and red lines represent the result with grid spacing of 10 and $40 \mu\text{m}$, respectively. Each result with grid spacing of 10 and $40 \mu\text{m}$ is shown in Figs. 8b and 8c, respectively, and it is an enlarged view near the downstream edge of the top electrode. We can observe in Fig. 5 the same characteristics in all cases: negative current spikes frequently appear in the negative-going voltage phase of the cycle, and positive current spikes with large amplitude appear in the positive-going voltage phase. Furthermore, Fig. 6 shows that the

Table 1 Electrical properties of dielectric materials

Material	Glass epoxy	Polyimide	PTFE
Relative permittivity	4.5–5.5	3.0	2.1
Dielectric breakdown strength [kV/mm]	22–27	18	16–19
Specific volume resistivity [Ωm]	10^{16} – 10^{18}	$>10^{19}$	$>10^{22}$

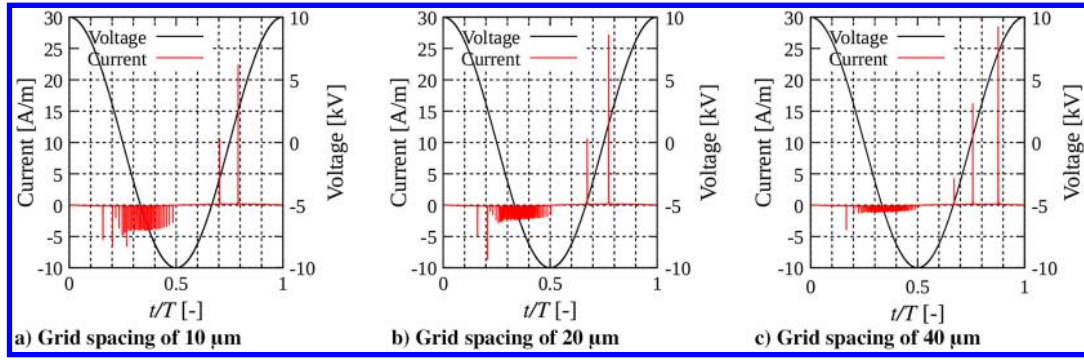


Fig. 5 Time history of voltage applied and discharge current waveform with different grid spacing.

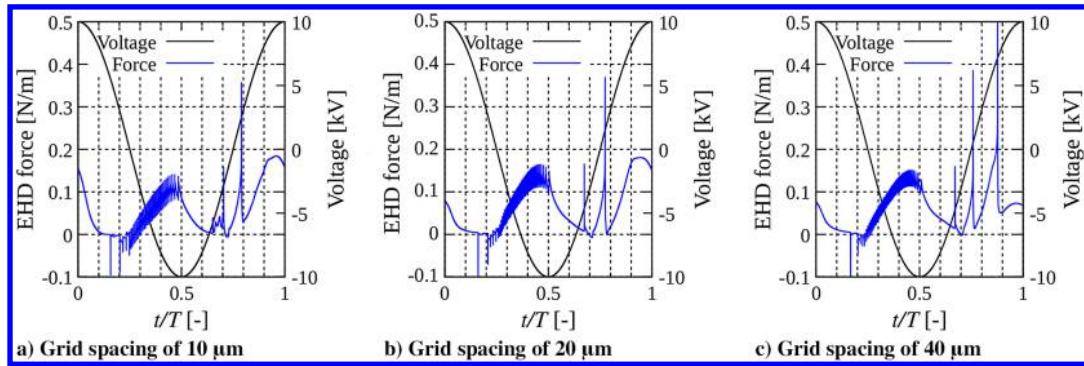


Fig. 6 Time history of voltage applied and EHD force with different grid spacing.

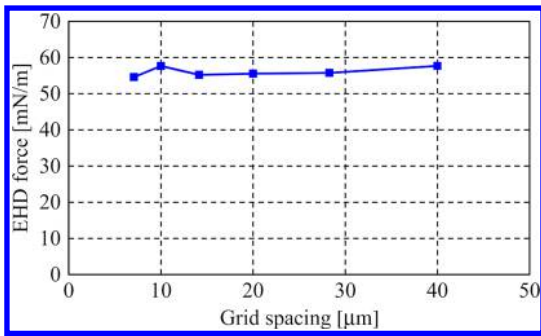


Fig. 7 Relation between the time-averaged EHD force and the computational grid spacing.

characteristics of EHD force generation are qualitatively same in all cases: the EHD force is generated when the discharge current spikes appear in both of the negative-going and positive-going voltage phase. On the other hand, the grid spacing has quantitative influence on the amplitude and frequency of the current spikes: the amplitude of negative current spikes increases and the number of positive current spikes decreases as the grid spacing decreases. However, as shown in Fig. 7, the influence of grid spacing on the magnitude of EHD force is quite small (less than 5%). In addition, as shown in Fig. 8a, the EHD force field extends higher in the vertical direction in the results with 40 μm and extends longer in the downstream in the results with 10 μm ; however, the area where strong EHD force is dominantly generated is not significantly affected by the grid spacing, as shown in Fig. 8b. Therefore, in terms of the time-averaged and

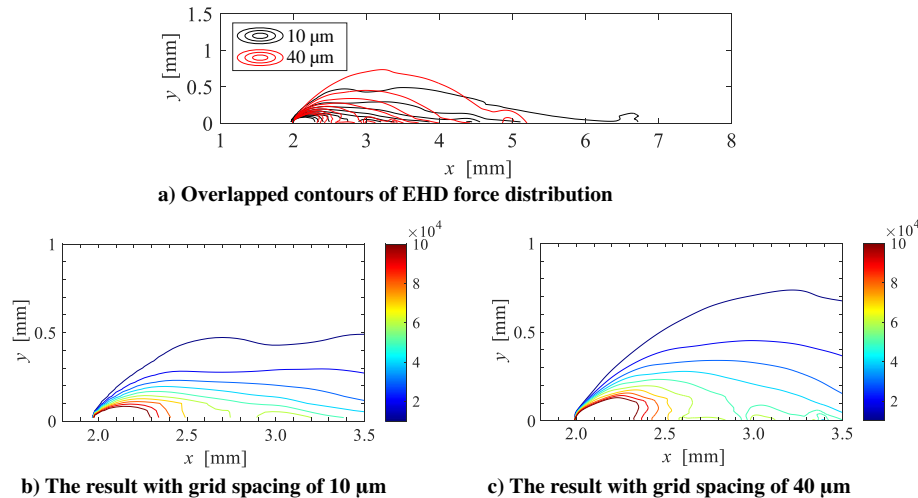


Fig. 8 Comparison of the time-averaged EHD force distribution between the results with grid spacing of 10 and 40 μm .

space-integrated value and distribution of the EHD force generation, we conclude that the grid convergence is achieved in practical range of grid spacing for the DBD plasma actuator.

B. Induced Jet Structure

In this section, we evaluate the reproducibility of the plasma simulation from the viewpoint of jet structure. Figures 9a and 9b show the time-averaged flowfields induced by the DBD plasma actuator from the simulation and experimental result, respectively:

the x -component, y -component, and absolute value of the flow velocity are shown in upper, middle, and lower figures, respectively, and the streamlines are illustrated in the lower figure. In Fig. 9, the downstream edge of the top electrode is set at the x -axis origin, and the locations and extent of the electrodes and dielectric are indicated in Fig. 9. In addition, the time-averaged velocity profiles are plotted in Figs. 10 and 11. Figure 10 shows the profiles of the x component along the y axis at $x = 5, 15$, and 25 mm, and Fig. 11 shows the profiles of the y component along the x axis at $y = 1, 3$, and 5 mm

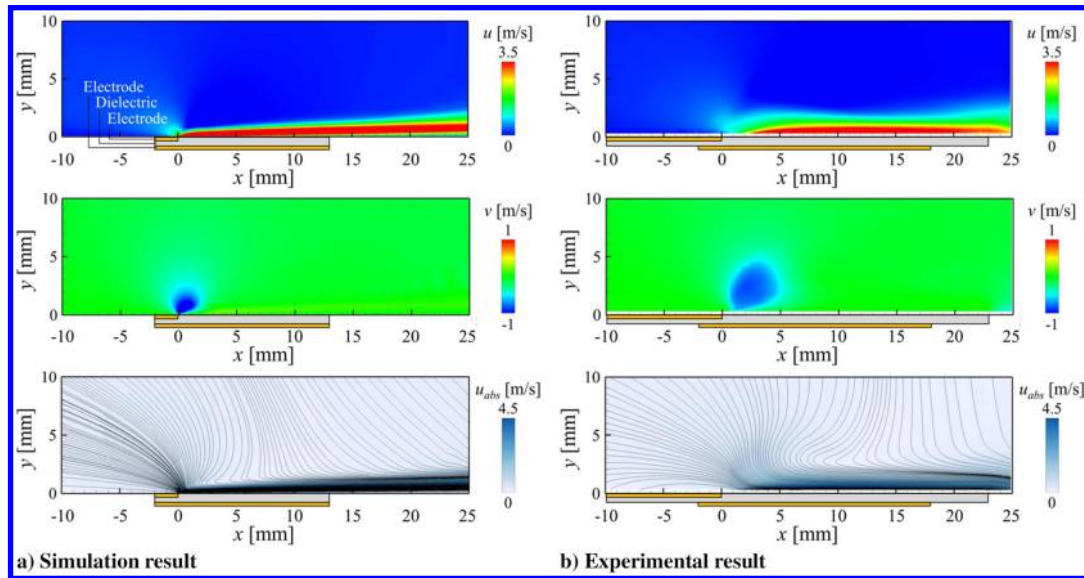


Fig. 9 Time-averaged flowfield: the x component of the velocity (top), the y component of the velocity (middle), and the absolute velocity and streamlines (bottom).

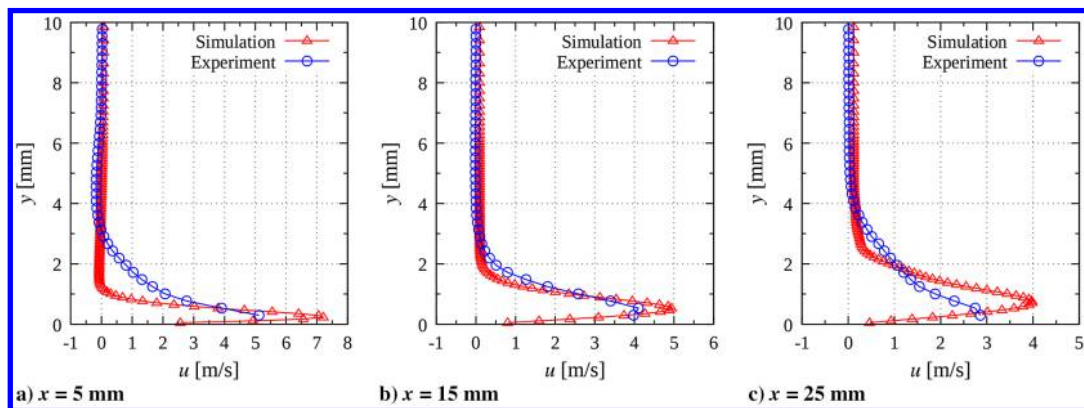


Fig. 10 Time-averaged profiles of the x component of flow velocity along the y axis.

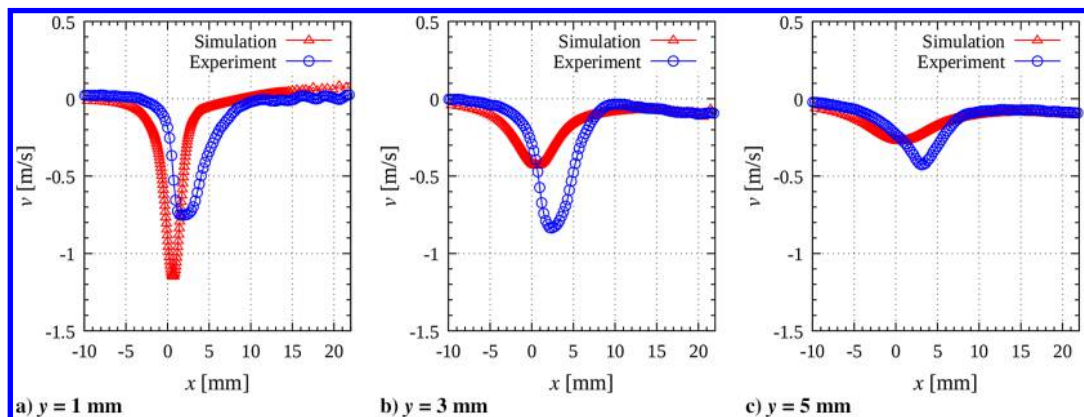


Fig. 11 Time-averaged profiles of the y component of flow velocity along the x axis.

above the surface. The voltage applied is the sinusoidal waveform voltage with an amplitude of 8 kV_p and a frequency of 5 kHz . The thickness of the dielectric layer is 0.5 mm . A glass epoxy is used as the dielectric material in the experiment, and a relative permittivity of the dielectric layer is set to 5.0 in the simulation.

Figures 9–11 demonstrate that the simulation successfully reproduces the qualitative characteristics of the flowfield induced by the DBD plasma actuator: as shown in Fig. 9, a wall jet is produced from the edge of the top electrode toward the positive x direction along the dielectric surface. The x -direction velocity decreases downstream from the edge of the top electrode, as shown in Fig. 10. In addition, the y -direction velocity field plotted in Fig. 9 highlights the flow in the negative y direction just downstream to the edge of the top electrode, that is, a fluid suction by the actuator. The magnitude of the negative y -direction velocity decays from $y = 1\text{ mm}$ toward $y = 5\text{ mm}$, as shown in Fig. 11. On the other hand, there are quantitative discrepancies between the simulation and experiment.

Comparing the x -direction velocity profiles between the simulation and experimental results plotted in Fig. 10, the jet induced is thinner in the simulation than in the experiments at all the positions of $x = 5, 15$, and 25 mm , as shown in Fig. 10. It is noted that the jet thickness is defined as the y coordinate of the position where the x -direction velocity decays to zero in this study. For instance, at $x = 5\text{ mm}$ position (Fig. 10a), the flow is induced in the area up to 1.25 mm height in the simulation, and on the other hand, up to 2.90 mm height in the experiment. This indicates that the plasma simulation underestimates the height (the range in the y direction) of EHD force distribution. In addition, Fig. 10 indicates that the maximum value of the x -direction velocity is larger in the simulation than in the experiment: the maximum velocity in the simulation is $7.21, 4.94$, and 3.98 m/s , and, on the other hand, $5.12, 4.10$, and 2.86 m/s in the experiment. This indicates that the simulation overestimates the strength of EHD force in the vicinity of the surface. Furthermore, comparing the y -direction velocity profiles plotted in Fig. 11, the position at which the velocity shows a negative maximum value is closer to the edge of the top electrode in the simulation than in the experiment: the position of maximum velocity is approximately $x = 0.7\text{ mm}$ in the simulation, and, on the other hand, in the experiment, $x = 1.9, 2.4$, and 3.1 mm at $y = 1, 3$, and 5 mm , respectively. This indicates that the simulation underestimates the width (the range in the x direction) of EHD force distribution, and as a result, a suction flow is induced in the area closer to the electrode edge in the simulation than in the experiment. Also, one can observe in Fig. 11 that the negative maximum value of the velocity at $y = 1\text{ mm}$ is larger in the simulation than in the experiment, whereas the negative maximum values at $y = 3$ and 5 mm are smaller. This is due to the underestimation of the height of EHD force distribution by the simulation, as described above. Furthermore, Fig. 9 shows that the high-velocity area is close to the downstream edge of the top electrode in the simulation; on the other hand, the jet seems to start downstream the electrode edge and is gradually accelerated in the experiment. In other words, the jet starts to be generated more upstream region and is more accelerated in the simulation. This indicates that the EHD force distribution extends more upstream toward the downstream edge of the top electrode and its strength is higher in the simulation. To be summarized, stronger EHD force is generated in a smaller area close to the edge of the top electrode in the simulation than in the experiment; consequently, stronger suction effect is caused in the area closer to the electrode edge and surface.

On the other hand, it should be noted that the discrepancy in the maximum value of the x -direction velocity is affected by the accuracy of the PIV measurement. As shown in Fig. 10, the boundary layer is not highly resolved, and the velocity at lower position than $y = 0.3\text{ mm}$ is not measured in the experiment. This is due to the lack of spatial resolution of the PIV measurement, lack of seeding particles in the vicinity of the surface, and the reflection of the laser light or the luminescence of the DBD plasma actuator. Accordingly, it seems that the origin and velocity of the induced jet near the downstream edge of the top electrode are not captured accurately in the experiment. Thus, for more accurate quantitative comparison of the maximum value of the x -direction velocity, especially near the

electrode edge, the visualization of flowfield using higher-resolution PIV measurements is required. Nonomura et al. [39] have reported that the maximum value of the x -direction velocity is higher in high-spatial-resolution PIV measurement than that in the conventional PIV measurement, and the conventional PIV measurement underestimates the strength of the EHD force near the downstream edge of the top electrode. On the other hand, the assumption of laminar flow in the simulation is valid within the measurement area and has little influence on the difference between the simulation and experiment. The Reynolds number is assumed to be too low for transition to turbulence because the flow velocity of induced jet is several m/s , as shown in Fig. 10, and the measurement area is in the vicinity of the electrode edge, where the EHD force is produced.

As discussed above, there are quantitative discrepancies between the simulation and experimental results in the thickness and maximum velocity of the jet induced; the maximum discrepancy in the jet thickness is 56% and that in the maximum value of the x -direction velocity is 29% . On the other hand, considering the quite simple numerical modeling, it can be concluded that the flowfield in the simulation is qualitatively in good agreement with that in the experiment, and the quantitative discrepancy is also not so large.

C. Voltage Amplitude Characteristics

The simulation and experiment are conducted for various applied voltage amplitude, and we qualitatively and quantitatively compare the simulation results with experimental results from the viewpoint of applied voltage amplitude characteristics of thrust force and power consumption, which are often used to evaluate the performance of a DBD plasma actuator. Figures 12 and 13 show the thrust force and power consumption as a function of the applied voltage amplitude, respectively. In Figs. 12 and 13, the simulation and experimental results are plotted together for comparison on log-log scale. The voltage applied is the sinusoidal waveform voltage with an amplitude of $5\text{--}10\text{ kV}_p$ and a frequency of 5 kHz . A 0.5-mm-thick glass epoxy (relative permittivity of 5.0), a 0.5-mm-thick polyimide (relative permittivity of 3.0), a 0.5-mm-thick PTFE (relative permittivity of 2.1), and a 1.0-mm-thick glass epoxy (relative permittivity of 5.0) are used as the dielectric layers in Figs. 12a–12d, respectively. Note that in the case of Fig. 12a, the voltage amplitude applied is limited to 9.5 kV_p due to dielectric breakdown in the experiment.

First, our experimental results are discussed by comparing to the experimental results reported in some previous studies. From the experimental results shown in Fig. 12 the thrust force increases as a power function of the voltage amplitude applied, and the slope of power-law line decreases as the voltage amplitude increases in all cases. Several straight lines with different slopes are plotted on Fig. 12 for reference. Although the values of power exponent are different depending on the conditions, they are in the range of 2.5 to 5 in all cases. The same characteristics have been reported in previous studies. Thomas et al. [40] have proposed 3.5 for low-voltage range and 2.3 for high-voltage range as the power exponent using a 6.35-mm-thick Teflon dielectric and a voltage frequency of 2 kHz . Ashpis and Laun [41] have reported that the thrust force increases as a power function of the voltage amplitude, and the power exponent reduces from about 4.8 to 2.5 as the voltage amplitude increased using a 0.35-mm-thick high-density polyethylene (HDPE) dielectric and a frequency of 1 kHz . It has been also reported that the value of power exponent depends on the conditions, such as a dielectric thickness and a voltage frequency: dielectric thicknesses of $1.5875, 3.175$, and 6.35 mm and a voltage frequency of $0.25\text{--}8\text{ kHz}$ was used by Ashpis and Laun [41].

In Fig. 13, the experimental result shows that the power consumption also increases as a power function of the voltage amplitude applied, and the slope of the power-law line is approximately constant value. The power exponent is in the range of $2.8\text{--}3.6$ depending on the conditions in this study. The same characteristics were observed in the previous studies. Enloe et al. [42] reported that the power consumption increased with the voltage amplitude to the power of 3.35 using a polyimide dielectric and a voltage frequency of 3 kHz . Murphy and Lavoie [43] proposed $3.2\text{--}3.36$ as the power

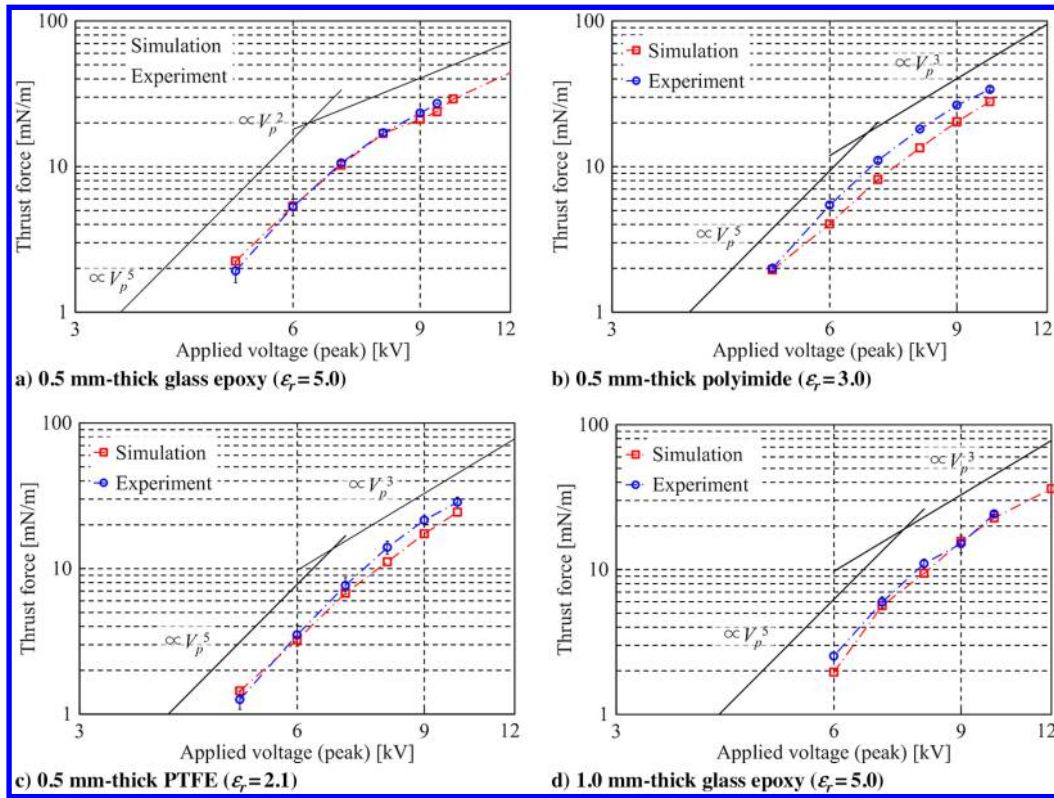


Fig. 12 Thrust force against voltage amplitude, V_p , on log-log scale for different actuator configurations.

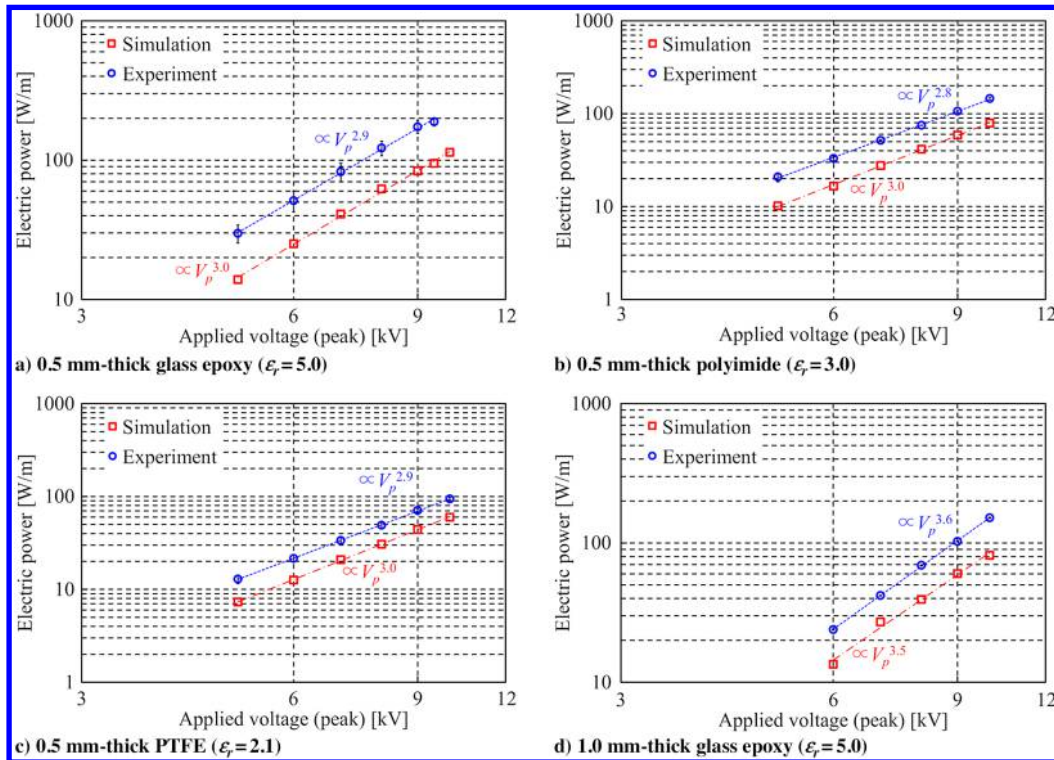


Fig. 13 Power consumption against voltage amplitude, V_p , on log-log scale for different actuator configurations.

exponent using a 0.356-mm-thick polyimide dielectric and a frequency of 4 kHz.

Although there are quantitative discrepancies in the power exponent among the reports, which are mainly caused by differences in the conditions as presented by Ashpis and Laun [41], the behaviors of the thrust force and power consumption in our experiments agree well with the observations reported in previous studies.

Next, we discuss the simulation results by comparing with the experimental results. We can clearly observe in Fig. 12 that the simulation results of thrust force have the same tendency as the experimental results as discussed above. Furthermore, the values of power exponent are in the range of 2.5–5 as in the experimental results. Comparing the amplitude of thrust force, the simulation tends to underestimate the thrust force compared with the experiment. The

maximum value of discrepancies, which are defined as relative difference of simulation result to experimental result, is +26.8% at 5 kV_p in the case of Fig. 12a, -22.3% at 8 kV_p in the case of Fig. 12b, +22.2% at 5 kV_p in the case of Fig. 12c, and -14.2% at 8 kV_p in the case of Fig. 12d. Note that, in the case of Figs. 12a and 12c, the amplitude of thrust force at 5 kV_p is quite small, which results in relatively large difference of over 20%. The discrepancies are less than 10% when the voltage is higher than 5 kV_p, and the simulation results coincide with the experimental results within the error bars in almost all cases (including 5 kV_p) of Figs. 12a and 12c. On the other hand, in the case of Fig. 12b, the discrepancy is relatively large in any cases: the averaged discrepancy is -16.7%. This is related to reproducibility of dielectric material characteristics (which will be discussed in Sec. IV.E, in detail). Therefore, we conclude that the simulation model has high capability to reproduce the voltage amplitude characteristics of the thrust force within 10–20%. Figure 13 shows that the simulation can reproduce qualitatively the voltage characteristics of power consumption observed in the experimental results. From quantitative comparison, although the values of power exponent agree well with the experimental results, the amplitudes of power consumption in the simulation are quite small compared with the experiment. The maximum value of discrepancies in each case is -53.5% at 5 kV_p in the case of Fig. 13a, -51.2% at 5 kV_p in the case of Fig. 13b, -43.4% at 5 kV_p in the case of Fig. 13c, and -46.3% at 10 kV_p in the case of Fig. 13d, respectively. In contrast to the thrust force, the discrepancies are similar values regardless of the voltage amplitude, and the averaged values are -50.9% in the case of Fig. 13a, -47.0% in the case of Fig. 13b, -39.1% in the case of Fig. 13c, and -42.0% in the case of Fig. 13d, respectively. Thus, the simulation underestimates the power consumption.

To provide further insight into the reproducibility of the thrust force and power consumption, the characteristics of discharge current waveform are compared between the simulation and experimental results. Figures 14 and 15 show the discharge current waveforms from the simulation and experimental results, respectively: the results plotted in Figs. 14 and 15 correspond to that plotted in Figs. 12a and 13a. As shown in Fig. 15, in both experimental results at lower

and higher voltages, negative current spikes, and positive current spikes frequently appear in the negative-going voltage phase and positive-going voltage phase of the cycle, respectively. On the other hand, Fig. 14 shows that although the simulation results have the same tendency in the negative-going voltage phase as the experimental results, the positive current spikes in the positive-going voltage phase do not appear at lower voltage (Fig. 14a) and only a few spikes appear even at higher voltage (Fig. 14b) in the simulation. This indicates that the characteristics of discharge current waveform in the simulation are qualitatively different from those in the experiment under the same conditions. On the other hand, by further increasing the voltage amplitude to 20 kV_p in the simulation, as shown in Fig. 14c, the positive current spikes in the positive-going voltage phase frequently appear as well as the experimental results; that is, the characteristics of discharge current waveform in the simulation qualitatively agree with that in the experiment. This implies that the threshold value of voltage amplitude for the current spike occurrence in the positive-going voltage phase is higher in the simulation than in the experiment. In addition, we can clearly observe in Figs. 14 and 15 that the number of current spikes is quite smaller in the simulation than that in the experiment. The underestimation in power consumption seems to be due to this difference.

The discrepancies in the threshold voltage and number of occurrences of positive current spikes are due to spanwise nonuniformity in discharge plasma structure. In the experiment, the local intensity of electric field at the electrode edge becomes higher because of stronger electric field concentration, which is caused from minute roughness in the electrode edge shape, resulting in occurrence of spot breakdown and a number of microdischarges in the spanwise direction [44]. The occurrence of one microdischarge corresponds to one spike in the discharge current waveform. On the other hand, in the 2D simulation, in which the spanwise nonuniformity in discharge plasma is not taken into account, only a single microdischarge occurs in the spanwise direction, resulting in higher threshold voltage and fewer occurrences of current spikes. To improve the quantitative reproducibility of power consumption, it is necessary to conduct three-dimensional plasma simulation considering the spanwise nonuniformity. Nishida et al. reported that the number of current spikes in the 3D

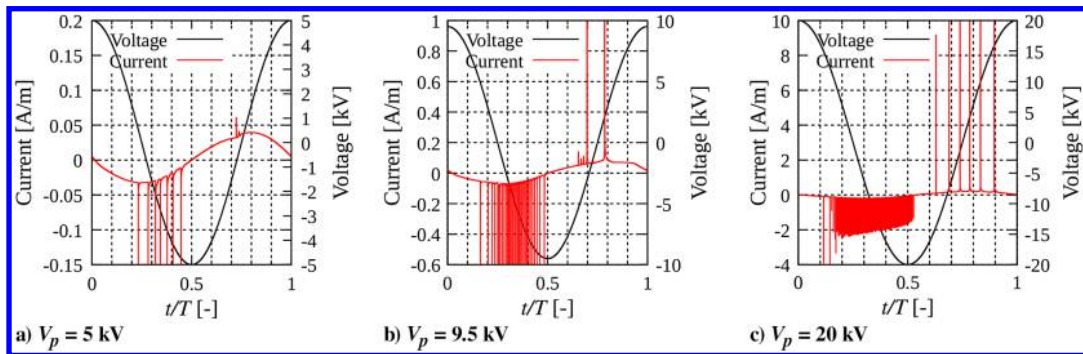


Fig. 14 Time history of voltage applied and discharge current waveform from simulation result.

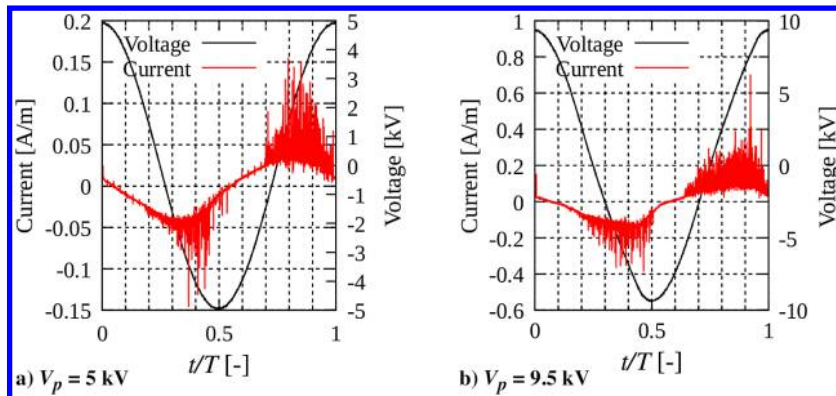


Fig. 15 Time history of voltage applied and discharge current waveform from experimental result.

simulation is much higher than that in the 2D simulation [33]. Thus, it cannot be assumed that the 2D simulation results match quantitatively the experimental results from a logical perspective; however, our results demonstrate that the thrust force in the 2D simulation matches that in the experiment well. The good agreement of the simulation with the experiment implies that the 2D simulation successfully reproduces the mean value of plasma density and electric field intensity under the condition adopted in this study. Nishida et al. also reported that the EHD force distribution obtained in the 2D simulation corresponds well to that in the 3D simulation from the viewpoint of time averaging and spanwise averaging [33]. It also supports our results that the thrust force in the simulation corresponds to that in the experiment only within 10–20% even though there are discrepancies in the discharge current waveform between the simulation and experiment. Note, however, that it is considered that the quantitative discrepancy is significantly affected by the conditions.

From the discussion in this section, the simulation can qualitatively reproduce the voltage amplitude characteristics of thrust force and power consumption. In regard to the quantitative reproducibility, although the simulation tends to underestimate the thrust force, the discrepancy is only within 10–20% depending on the dielectric material. Note that, as discussed in Sec. IV.B, the jet thickness is underestimated in the simulation, resulting in large loss due to the viscous drag on the surface compared with that in the experiment. Based on these results, the simulation is expected to reproduce the amplitude of EHD force generation within less than 10–20%. On the other hand, the simulation underestimates the values of power consumption by 40–50%. It is mainly due to the spanwise nonuniformity in discharge plasma structure, which also results in the discrepancy in characteristics of discharge current waveform.

D. Voltage Waveform Characteristics

Some previous studies have reported that the applied voltage waveform affects the performance of DBD plasma actuators significantly [45,46]. In this section, to investigate the reproducibility of the applied voltage waveform characteristics using plasma simulation, the applied voltage waveform characteristics of thrust force and power consumption are compared between the simulation and experiment qualitatively. Three typical types of waveforms are investigated: sinusoidal, triangular, and square waveforms. A 0.5-mm-thick glass epoxy (relative permittivity 5.0) is used as a dielectric layer. The voltage amplitude is 10 kV_p and the frequency is 750 Hz. The voltage frequency is set to relatively low value in order to reduce the influence of difference in the input voltage signal between the simulation and experiment. The simulation can reproduce accurately the discontinuous input voltage signal, such as the square waveform; however, in the experiment, the discontinuous signal is not accurately generated, and a maximum value of voltage gradient is limited by the maximum through rate of the high voltage amplifier (280 V/μs).

Figures 16 and 17 show the experimental and simulation results, respectively. In Figs. 16 and 17, the voltage waveform characteristics of thrust force and power consumption are plotted in upper and lower graphs, respectively. A comparison between Figs. 16 and 17 indicates that the voltage waveform characteristics of thrust

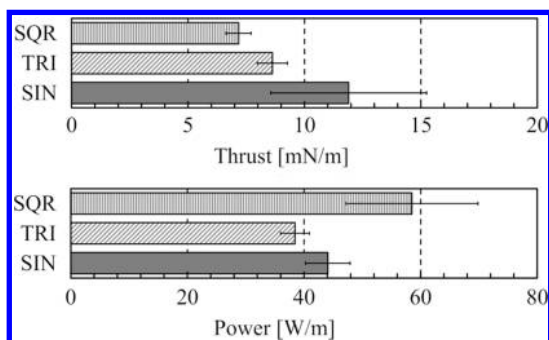


Fig. 16 Experimental results of voltage waveform characteristics of thrust force and power consumption in the case of 10 kV_p voltage amplitude and 750 Hz frequency.

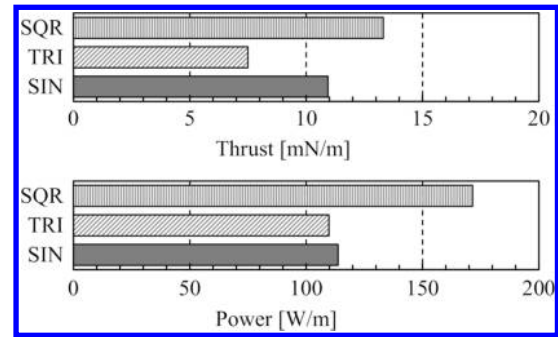


Fig. 17 Simulation results of voltage waveform characteristics of thrust force and power consumption in the case of 10 kV_p voltage amplitude and 750 Hz frequency.

force are qualitatively different between the experiment and simulation. The sinusoidal waveform produces larger force than the triangular waveform in both the experiment and simulation, and especially, the sinusoidal waveform produces the largest force in the experiment. However, the square waveform produces the largest force in the simulation. From the quantitative viewpoint, the relative difference of simulation result to experimental result is small: −8% and −13% in the case of the triangular and sinusoidal waveforms, respectively; however, there is a significant difference in the case of the square waveform: the relative difference is +86%. On the other hand, the simulation reproduces the qualitative waveform characteristics of power consumption: the power consumption is the smallest in the case of the triangular waveform, and it is the largest in the case of the square waveform. However, from the quantitative viewpoint, there is quite large discrepancy between the experiment and simulation.

Figures 18 and 19 show the discharge current waveforms from the experimental and simulation results, respectively. In the experiment, a number of positive current spikes appear in the positive-going voltage phase in all cases of three waveforms, as shown in Fig. 18. On the other hand, in the simulation, a few current spikes in the positive-going voltage phase appear only in the case of square waveform, as shown in Fig. 19; that is, the characteristics of discharge current waveform in the simulation are different from those in the experiment. However, by adjusting the conditions in the simulation as discussed in Sec. IV.C, the simulation becomes to be able to qualitatively reproduce the characteristics of discharge current waveform: the simulation results in the case that the voltage amplitude is increased to 20 kV_p and the frequency is increased to 5 kHz are shown in Figs. 20 and 21. Figures 19 and 20 plot the thrust force and power consumption, and Fig. 21 plots the discharge current waveform. Note that the voltage slope of square waveform is changed so that the shape of square waveform in the case of 20 kV_p amplitude and 5 kHz frequency is similar to that in the case of 10 kV_p amplitude and 750 Hz frequency. We can observe in Fig. 21 that a number of current spikes appear in the positive-going voltage phase in all cases of three waveforms. In addition, a comparison between Figs. 16 and 20 shows that the simulation can reproduce the qualitative waveform characteristics of not only power consumption, but also thrust force.

From the discussion above, the simulation can reproduce the qualitative waveform characteristics of power consumption, regardless of the characteristics of discharge current waveform. On the other hand, to reproduce the voltage waveform characteristics of thrust force, it is necessary that the characteristics of discharge current waveform in the simulation are qualitatively consistent with those in the experiment.

Note that the square waveform has been reported to provide the largest thrust force in the previous studies [45,46]; that is, the voltage waveform characteristics of thrust force in our experiment are in disagreement with previous studies. This is because the voltage waveform characteristics vary depending on the conditions, such as the voltage amplitude, dielectric thickness, and material, as reported in Ref. [47].

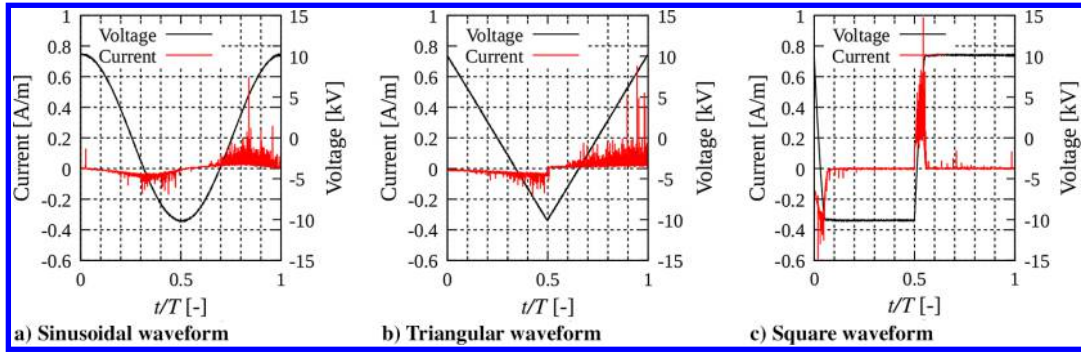


Fig. 18 Experimental results of time histories of voltage applied and discharge current waveform in the case of 10 kV_p voltage amplitude and 750 Hz frequency.

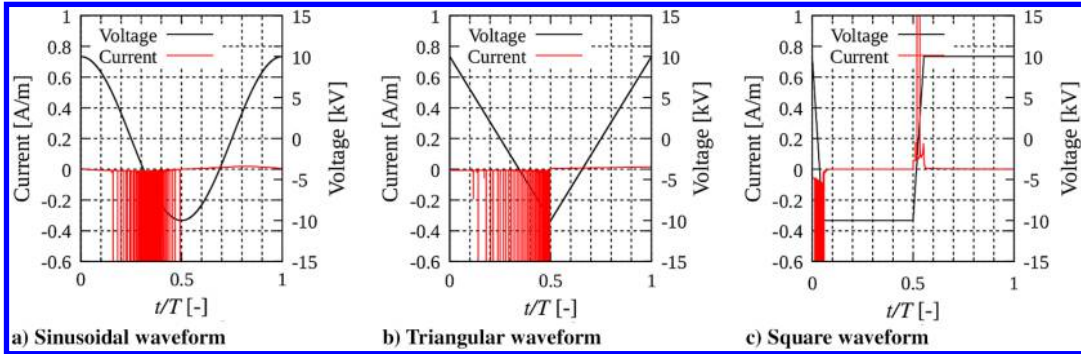


Fig. 19 Simulation results of time histories of voltage applied and discharge current waveform in the case of 10 kV_p voltage amplitude and 750 Hz frequency.

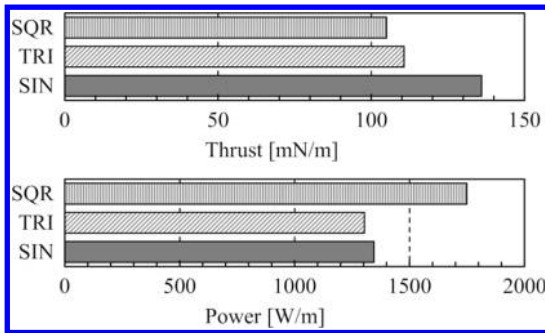


Fig. 20 Simulation results of voltage waveform characteristics of thrust force and power consumption in the case of 20 kV_p voltage amplitude and 5 kHz frequency.

E. Dielectric Material Characteristics

Some previous studies have reported that dielectric material also has the influence on the performance of DBD plasma actuators [3]. In this section, to investigate the reproducibility of dielectric material

characteristics, the dielectric material characteristics of thrust force and power consumption are qualitatively compared between the simulation and experiment. Three types of materials are used as the dielectric: glass epoxy, polyimide, and PTFE, whose relative permittivity is 5.0, 3.0, and 2.1, respectively. Because the permittivity is considered as one of the parameters contributing to the dielectric material characteristics, the experimental results are compared with the simulation results in which the relative permittivity is set to the same value as that of the material used in the experiment. Therefore, the simulation results represent the permittivity characteristics in a strict sense, whereas the experimental results represent the dielectric material characteristics. The voltage applied is a sinusoidal voltage with 10 kV_p amplitude and 5 kHz frequency, and the dielectric thickness is 0.5 mm.

Figures 22 and 23 show the experimental and simulation results, respectively. In Figs. 22 and 23, the dielectric material characteristics of thrust force and power consumption are plotted in upper and lower graphs, respectively. A comparison between Figs. 22 and 23 indicates that the dielectric material characteristics of thrust force are qualitatively different between the experiment and simulation: although the thrust force increases with increasing the permittivity in the

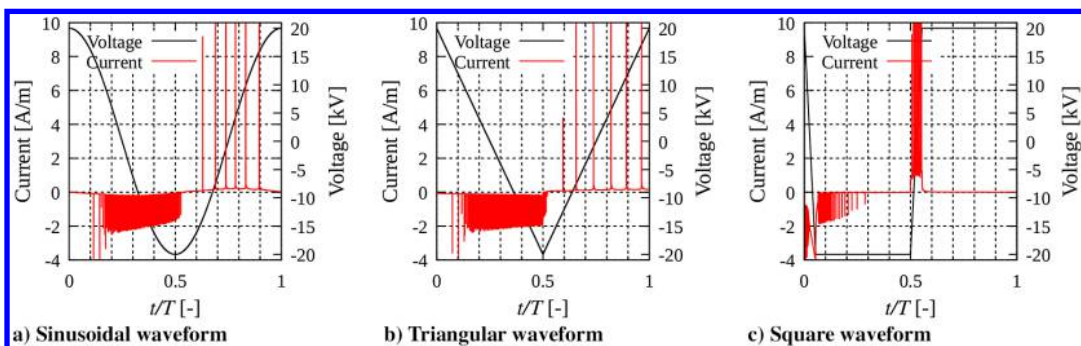


Fig. 21 Simulation results of time histories of voltage applied and discharge current waveform in the case of 20 kV_p voltage amplitude and 5 kHz frequency.

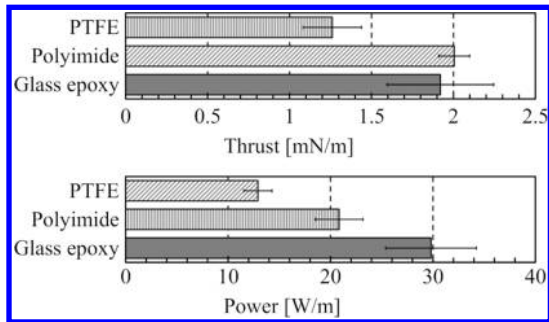


Fig. 22 Experimental results of dielectric material characteristics of thrust force and power consumption in the case of 5 kV_p voltage amplitude and 5 kHz frequency.

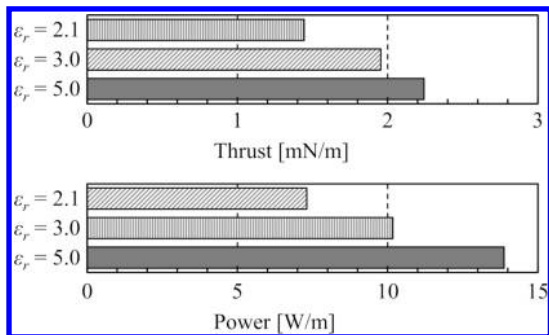


Fig. 23 Simulation results of permittivity characteristics of thrust force and power consumption in the case of 5 kV_p voltage amplitude and 5 kHz frequency.

simulation, the thrust force in the case of polyimide is larger than that in the case of glass epoxy in the experiment. Note that the quantitative discrepancy between the experiment and simulation is only within 10–20%. On the other hand, in regard to the power consumption, although there is quite large quantitative discrepancy between the

experiment and simulation, the simulation can reproduce the qualitative characteristics: the power consumption increases with the increase in the permittivity.

Figures 24 and 25 show the discharge current waveforms from the experimental and simulation results, respectively. From the comparison between Figs. 24 and 25, the characteristics of discharge current waveform in the simulation are qualitatively different from those in the experiment in terms of current spikes in the positive-going voltage phase. On the other hand, the simulation can qualitatively reproduce the characteristics by adjusting the conditions, as described in previous section: the simulation results in the case of 20 kV_p voltage amplitude and 10 kHz frequency are shown in Figs. 26 and 27. However, the comparison between Figs. 22 and 26 shows that the dielectric material characteristics of thrust force in the simulation are different from those in the experiment: the thrust force is almost the same among three permittivity in the simulation. Thus, even in the case that the characteristics of discharge current waveform in the simulation are consistent with those in the experiment qualitatively, the simulation cannot reproduce the dielectric material characteristics of thrust force.

This discrepancy between the experiment and simulation indicates that other parameters affect the thrust force characteristics. One of the parameters characterizing the dielectric material is the secondary electron emission coefficient. We have confirmed that the secondary electron emission coefficient affects the characteristics of discharge current waveform: as the secondary electron emission coefficient decreases from 0.1 to 0.01, the number of current spikes in the positive-going voltage phase increases, and the strength of EHD force reduces. In addition, it has been reported that the thrust force differs even when the permittivity is almost the same value in experimental study [47]. Therefore, to reproduce the dielectric material characteristics of thrust force, not only permittivity but also other dielectric parameters have to be set appropriately in the simulation.

F. Discussion

The results shown in Secs. IV.C–IV.E reveal that the characteristics of discharge current waveform affect the reproducibility of the simulation. In this section, to clarify the mechanism, we investigate the

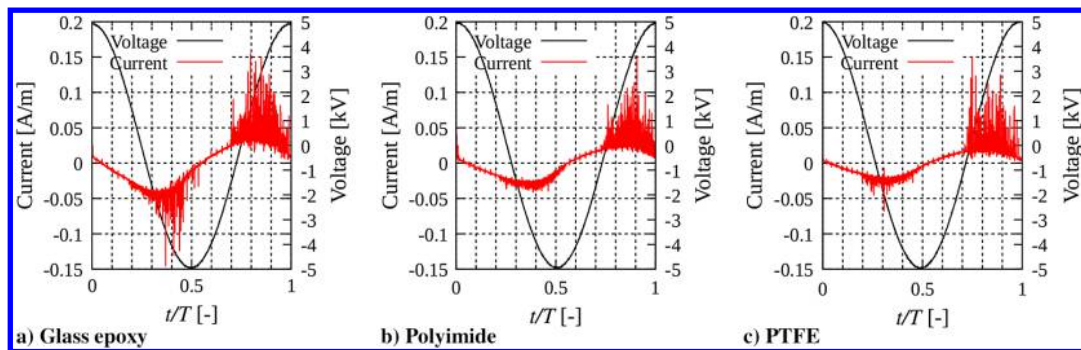


Fig. 24 Experimental results of time histories of voltage applied and discharge current waveform in the case of 5 kV_p voltage amplitude and 5 kHz frequency.

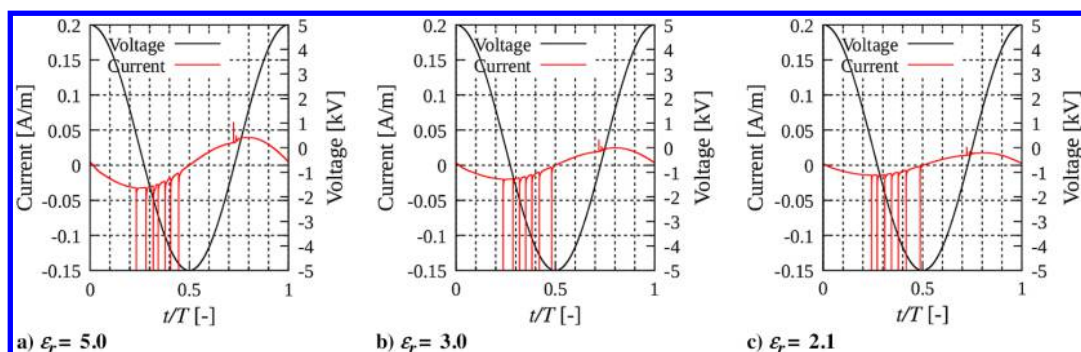


Fig. 25 Simulation results of time histories of voltage applied and discharge current waveform in the case of 5 kV_p voltage amplitude and 5 kHz frequency.

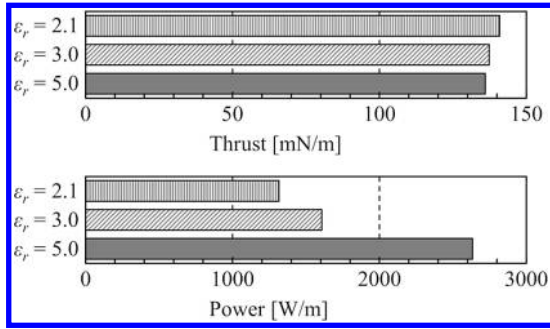


Fig. 26 Simulation results of permittivity characteristics of thrust force and power consumption in the case of 20 kV_p voltage amplitude and 10 kHz frequency.

discharge mode in a DBD plasma actuator, and then discuss the effect of discharge mode on EHD force generation characteristics.

The simulation results demonstrate that the discharge mode is totally different between the negative-going and positive-going voltage phase of the cycle. The discharge in the negative-going voltage phase is categorized as the glow-type discharge, regardless of the conditions. Figure 14 indicates that although the peak value and number of occurrences of current spikes in the negative-going voltage phase increase with the increase in the voltage amplitude, the characteristics do not change qualitatively. Regarding the discharge plasma structure, the discharge plasma forms around the exposed electrode edge and expands along the surface with time. On the other hand, in the positive-going voltage phase, the discharge mode transition occurs depending on the conditions. In the case of lower voltage amplitude and frequency, the positive current spike does not appear, as shown in Fig. 14a, and a diffusive discharge plasma forms continuously near the exposed electrode edge. This discharge is categorized into the corona-type discharge. On the other hand, in the case of higher voltage amplitude and frequency, large current spikes periodically appear, as shown in Fig. 14c. When the current spike occurs, a high-density plasma column propagates away from the exposed electrode along the surface. This discharge is categorized into the streamer-type discharge.

The EHD force generation characteristics are strongly affected by the discharge mode transition. Figures 28a and 28b show the time histories of space integrated EHD force and voltage waveform in the cases of 5 and 20 kV_p corresponding to the cases shown in Figs. 14a and 14c, respectively. Comparing between Figs. 28a and 28b, the EHD force generation characteristics during the negative-going voltage phase are qualitatively the same in both cases: the EHD force increases during the glow-type discharge, and then, it reduces. On the other hand, the EHD force characteristics during the positive-going phase are totally different between the cases of corona-type and streamer-type discharges. In the case of the corona-type discharge, the EHD force continuously increases with time, as shown in Fig. 28a. In contrast, in the case of the streamer-type discharge, the EHD force generation is disturbed by the streamer discharge occurrence because the EHD force is quasi zero in the channel of streamers (quasi-neutral plasma), as shown in Fig. 28b. As a result, as clearly observed in Fig. 28, the EHD force generation during the positive-going voltage phase is dominant in the case that the corona-type discharge occurs in the positive-going voltage phase. On the other hand, the EHD force generation during the negative-going voltage phase is dominant in the case that the streamer-type discharge occurs in the positive-going voltage phase. Enloe et al. conducted the time-resolved thrust measurement varying the reduced electric field strength of plasma actuator [48]. They reported that the positive-going voltage phase generates higher thrust under low reduced electric field condition (i.e., low applied voltage or high ambient pressure), and the negative-going voltage phase becomes to generate higher thrust under high reduced electric field condition. Their experimental results support our simulation results. In conclusion, for reproduction of the detailed characteristics of thrust force, such as the voltage waveform characteristics, the discharge mode in the simulation needs to be consistent with that in the experiment.

Based on the above discussion, the discrepancy in the discharge mode between the simulation and experiment is considered to contribute to the significant quantitative difference in thrust force in the case of the square waveform, as stated in Sec. IV.D. The square waveform has a period with a quite steep voltage slope, compared with the sinusoidal and triangle waveforms. In the experiment, the streamer-type discharge repetitively occurs at high frequency during

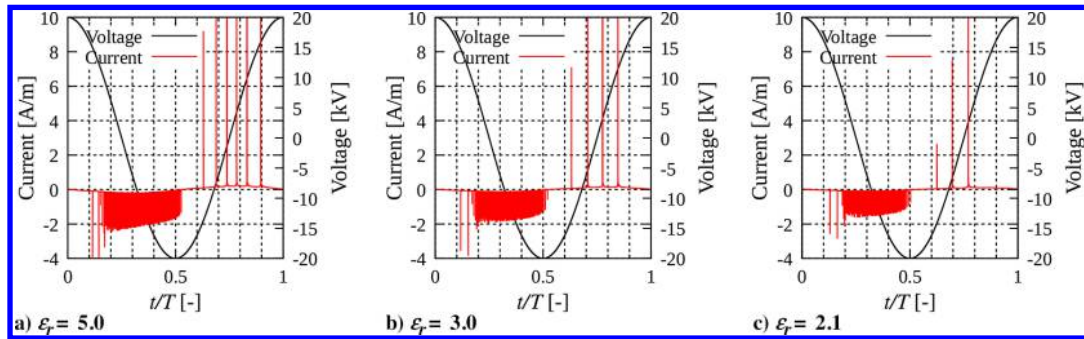


Fig. 27 Simulation results of time histories of voltage applied and discharge current waveform in the case of 20 kV_p voltage amplitude and 10 kHz frequency.

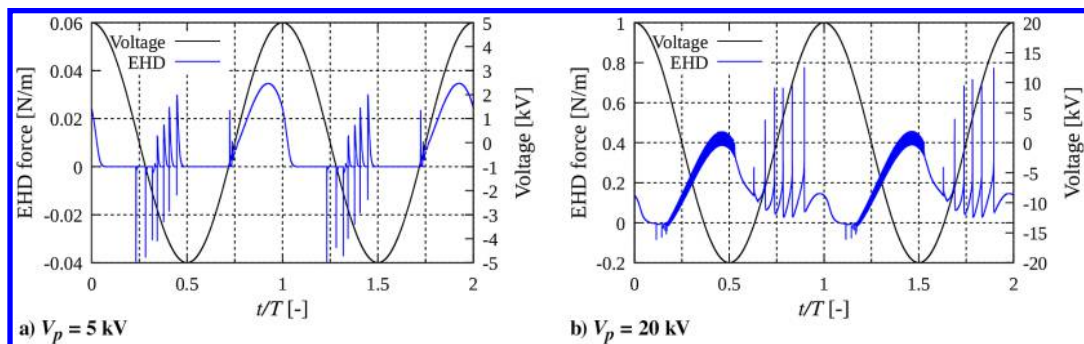


Fig. 28 Time history of voltage applied and EHD force generation.

the short period of steep voltage slope, and dense plasma is generated. The electric field is presumed to be eliminated because the dielectric surface gets charged by the plasma, and as a result, in a long period with a constant voltage after the steep slope, the discharge is less likely to be maintained in the experiment. On the other hand, in the simulation, the corona-type discharge continuously occurs and the EHD force is generated in the constant voltage period. Thus, the influence of the discrepancy in the discharge mode seems to make noticeable differences especially in the case of the square waveform, which has the quite steep voltage slope.

Finally, in terms of the quantitative reproducibility of the modeling, the secondary electron emission coefficient has to be carefully set, as stated previously. However, the secondary electron emission coefficient is difficult to be set physically. There are few measurement results of the coefficient for the materials typically used as the dielectric in the DBD plasma actuators. Therefore, we suggest using the secondary electron emission coefficient as a tuning parameter to adjust the values of thrust force and the threshold voltage of discharge mode transition. Note that, because the qualitative characteristics of discharge are not affected by the secondary electron emission coefficient, the simulation using the typical value ($\gamma = 0.05$) is capable of evaluating the physics in the DBD plasma actuator, except for the dielectric material characteristics of thrust force. Furthermore, the typical value of recombination coefficient is considered to be suitable for the quantitative reproduction of thrust force, while the recombination processes are considered quite simply. On the other hand, considering chemical reactions in more detail could improve the quantitative reproducibility of thrust force and discharge mode transition, in addition to considering the spanwise nonuniformity. It is necessary to investigate the quantitative influence by considering detailed chemical reactions, such as detailed recombination and photoionization as future works.

V. Conclusions

In this study, discharge plasma behavior of the DBD plasma actuator was simulated based on the simple fluid model, in which the electron, one type of positive ion, and one type of negative ion were taken into account. To validate the numerical modeling of discharge plasma, the simulation and experiment were conducted under the same conditions.

First, the flowfield obtained in the simulation was compared with that obtained using the PIV measurement. Although the simulation underestimated the jet thickness and overestimated the maximum flow velocity, it is concluded that the flowfield in the simulation was qualitatively in good agreement with that in the experiment, and the quantitative discrepancy was not so large, considering the quite simple numerical modeling.

Next, the reproducibility of thrust force and power consumption were investigated. In regard to the quantitative reproducibility, the values of thrust force in the simulation were consistent with those in the experiment within 10–20%. On the other hand, the values of power consumption were smaller in the simulation than those in the experiment by 40–50%. This is mainly because the 2D simulation cannot reproduce a number of microdischarges occurring in the spanwise direction. In regard to the qualitative reproducibility, although the voltage amplitude, voltage waveform, and dielectric material characteristics of power consumption in the simulation agree with those in the experiment, the voltage waveform and dielectric material characteristics of thrust force were different between the simulation and experiment.

The discrepancies in the characteristics of thrust force are due to the discharge mode transition from the corona-type to the streamer-type discharge. The simulation results revealed that the discharge mode largely affected the EHD force generation characteristics. To reproduce the voltage waveform characteristics of thrust force, the discharge mode in the simulation needs to be consistent with that in the experiment. In addition, the dielectric parameters, such as permittivity and secondary electron emission coefficient, need to be set to appropriate values for each material used in the experiment in order to reproduce the dielectric material characteristics of thrust force.

As future works, for more accurate quantitative comparison of the maximum velocity of the induced jet, especially near the electrode edge and dielectric surface, the visualization of flowfield using high-spatial-resolution PIV measurements will be conducted. In addition, this study focuses on the time-averaged characteristics of the DBD plasma actuator aiming to validate the three-fluid model from the viewpoint of total momentum input and time-averaged power consumption. On the other hand, the detailed characteristics of spatial and temporal distribution of EHD force become more critical when unsteady pulsed actuations are used for flow control. The validation in terms of time-variation characteristics of the DBD plasma actuator is one of the important issues in future works. Furthermore, to improve the quantitative reproducibility, it is necessary to investigate the influence by considering chemical reactions in more detail, in addition to considering the spanwise nonuniformity.

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